

How Well Does It Generalize?

A Robustness Study of Hausa Machine-Generated Text Detection

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Abstract

Sani et al. (2025) introduced the first dedicated Hausa machine-generated text (MGT) detector, an AfroXLMR-large classifier trained on 2,586 paired articles from seven Hausa news outlets and Gemini 2.0 Flash, reporting F1 of 0.9921 on a headline-grouped split. They identified three limitations of their work: a single news domain, a small batch and short training schedule, and a single generator (Gemini 2.0 Flash) for the machine-generated data. We extend their work along the third axis. First, we replicate their reported result across four random seeds, obtaining F1 of 0.992 ± 0.005 . Test recall on the AI class is 1.000 at every seed. Second, we analyse the detector’s false positives. All errors fall on Human articles, and two native-speaker reviewers confirm these are genuine, natural Hausa journalism. The clearest correlate of misclassification is the absence of numerical content: the median digit fraction is 0.0% in misclassified articles vs. 0.6% in correctly classified ones. Third, we test the detector on six non-training generators spanning four model families. Detection rates span from 0% (Aya-Expanse 8B) to nearly 100% (Gemma 4). All three Google-family models in our experiment (training-distribution Gemini 2.0 Flash and the two non-training Gemma 4 variants) are detected at rates above 98% with mean $P(\text{machine})$ above 0.98. No non-Google generator exceeds 21%. Three contamination checks fail to explain this away: generated articles overlap very little with their human sources, detection barely moves when we remove every article that shares a 13-gram with its source, and the Google output that gets detected is at least as diverse as everyone else’s, so it is not memorised or templated text. What the detector has learned looks less like a general Hausa MGT signal and more like a Gemini-family stylistic fingerprint. We discuss what this means for deployment, including a 3.1% false-positive rate on real Hausa journalists.

1 Introduction

Machine-generated text detection has become a pressing concern for content moderation, academic integrity, and combating misinformation. Most published detectors and benchmarks focus on English. Hausa, a Chadic language with over 100 million speakers across West Africa (Inuwa-Dutse, 2021) and a major media language in Nigeria, has only recently received attention.

The imbalance matters beyond coverage. Detection tools shape who gets believed. A detector that misfires on genuine human writing casts suspicion on real journalists; one that misses machine text lets fabricated news circulate. Hausa is widely spoken but thinly resourced, and independent fact-checking infrastructure is scarcer than for English, so the reliability of a single published detector carries unusual weight. Knowing when that detector can be trusted, and when it cannot, is part of what equitable access to trustworthy information means for the language.

Sani et al. (2025) released the first dedicated Hausa MGT detection benchmark at AfricaNLP 2025. The dataset pairs 2,586 articles scraped from seven Hausa news outlets with 2,586 articles generated by Gemini 2.0 Flash from the same headlines. They fine-tuned four Afri-centric pre-trained models on the resulting dataset, with AfroXLMR-large achieving the highest F1 at 0.9921. They split the data with no headline overlap across train/validation/test to avoid leakage, and acknowledged three limitations: single domain (news), short training schedule (three epochs, batch 8), and a single generator (Gemini 2.0 Flash) for the AI class.

This paper takes up the third limitation directly. The single-generator concern is the most consequential for deployment: a detector trained on one model’s output may fail on output from other generators in unpredictable ways. We also examine two

related questions that the original paper did not. We re-train the AfroXLMR-large detector across four random seeds to quantify run-to-run variance, and analyse the detector’s false positives to characterise what the model has learned.

We organise the study around three research questions:

RQ1: Does the reported AfroXLMR-large result replicate across random seeds?

RQ2: What kinds of Human articles does the detector falsely flag as machine-generated?

RQ3: Does a detector trained on Gemini 2.0 Flash transfer to non-training generators?

Our contributions are:

1. A multi-seed replication of [Sani et al. \(2025\)](#)’s baseline across four random seeds (mean F1 = 0.992 ± 0.005). The reported number sits at our mean, confirming the result is robust to initialisation. Recall on the AI class is 1.000 at all four seeds.
2. An error analysis showing that all detector errors are false positives on genuine human-written news (confirmed by two native-speaker reviewers). The strongest correlate of misclassification is missing numerical content. We expected the article’s register to matter, but native-speaker review did not bear that out.
3. A cross-generator evaluation across six non-training generators (Cohere Aya, Meta Llama 3.1, Alibaba Qwen 3, Google Gemma 4 E4B, Google Gemma 4 E2B, Meta NLLB-200 round-trip). Detection rates span from 0% to 100%. Both non-training Google-family models are detected at rates above 98%, matching the training generator’s 100% rate; no non-Google generator exceeds 21%. Our reading is that the detector has learned a Gemini-family stylistic signature, not a general MGT signal.

2 Related Work

2.1 MGT detection

Machine-generated text detection has matured since the release of GPT-3-class models. Zero-shot statistical approaches ([Gehrmann et al., 2019](#);

[Mitchell et al., 2023](#)) and supervised classifiers fine-tuned on paired human/AI corpora are the dominant families. Cross-generator transfer has been studied repeatedly in English. The RAID benchmark ([Dugan et al., 2024](#)) and the M4 multilingual benchmark ([Wang et al., 2024](#)) both find that supervised detectors trained on output from one generator fail to transfer to output from other generators. We extend this line of work to Hausa, where we find the failure is not uniform: transfer ranges from total to nonexistent depending on the generator’s family.

2.2 African-language MGT detection and resources

[Sani et al. \(2025\)](#) is the first dedicated Hausa MGT detection study to our knowledge. Related work on African-language NLP has focused on translation ([Team et al., 2022](#); [Adelani et al., 2022](#)), language identification, and general-purpose model development ([Üstün et al., 2024](#); [Dang et al., 2024](#)). The AfroXLMR family of models that powers the detector ([Alabi et al., 2022](#); [Adelani et al., 2024](#)) is adapted from XLM-R via multilingual adaptive fine-tuning on African languages, including Hausa. Robustness studies for African-language NLP systems remain rare.

2.3 Hausa orthography

Standard written Hausa uses four hooked consonants (ḃ, ḉ, ɓ, y’), which fall outside the ASCII range. These characters are commonly mishandled in NLP pipelines. We apply Unicode repair ([Speer, 2019](#)) to generator inputs in our cross-generator experiments to ensure non-Hausa-trained models receive valid orthography.

3 Methodology

3.1 Data

We use the [Sani et al. \(2025\)](#) release without modification. The dataset contains 5,172 articles across two label classes: 2,586 Human articles scraped from seven Hausa media outlets and 2,586 AI articles produced by Gemini 2.0 Flash from the corresponding headlines. We follow their headline-grouped 80/10/10 split: each headline is assigned to one of three splits, and both the Human and AI article for that headline go together.

The released dataset contains a class-correlated asymmetry in Hausa orthography. Human articles average 29.3 hooked-consonant occurrences per article after Unicode normalisation; Gemini articles

average 14.4 (a $2.0\times$ ratio). 72.7% of Human articles contain at least one hooked letter, vs. 37.9% of Gemini articles. The class composition is also subdomain-skewed: 53.5% politics, 10.4% health, 9.7% sports, with smaller shares for business, entertainment, religion, and technology (Sani et al., 2025).

3.2 Detector training

We follow Sani et al. (2025)’s hyperparameters: AfroXLMR-large with learning rate 1×10^{-5} , weight decay 0.01, maximum sequence length 512, and effective batch size 32 (per-device batch 8, gradient accumulation 4). We train for up to 8 epochs with early stopping on validation F1 (patience 2), an extension of their 3 epochs. The longer schedule is motivated by inspection of validation curves on seed 42: validation F1 was 0.961 at epoch 3 (matching the paper’s reported number) but peaked at 0.990 at epoch 6 before declining slightly. The original 3-epoch schedule appears to have stopped training short of the achievable peak, though the paper’s hyperparameters reach the reported test F1 within run-to-run variance regardless.

We train at fp32 due to a classifier-head initialisation issue in transformers 5.8.1 that produces NaN weights when the classification head is newly added. We re-initialise the head manually with $\mathcal{N}(0, 0.02)$ weights and zero biases. We run four random seeds (42, 1, 7, 123) and report mean and standard deviation across runs. The seed-42 checkpoint is used for all subsequent experiments unless noted.

3.3 Cross-generator data

For each of six non-training generators, we produce 260 Hausa articles paired with test-split Human articles. Each generator is given a chat-template prompt asking it to write a Hausa news article matching the topic of an input headline. We test:

- **Aya-Expanse 8B** (Dang et al., 2024): Cohere; explicit African-language training.
- **Llama 3.1 8B Instruct** (Grattafiori et al., 2024): Meta; English-centric.
- **Qwen 3 8B** (Yang et al., 2025): Alibaba; 119-language multilingual pre-training.
- **Gemma 4 E4B** (DeepMind, 2026): Google; 140-language multilingual.

- **Gemma 4 E2B** (DeepMind, 2026): smaller variant of the same family.
- **NLLB-200 distilled 600M** (Team et al., 2022): Meta; translation paradigm, Hausa $\rightarrow$ English $\rightarrow$ Hausa round-trip.

All generators run via vLLM with bf16, temperature 0.8, top-p 0.95. Headline inputs are passed through ftfy (Speer, 2019) to repair mojibake (Appendix B). For thinking-capable models (Qwen 3, Gemma 4), thinking mode is explicitly disabled via the chat template so the full output-token budget is available for visible output.

The NLLB round-trip serves a second purpose. Because it preserves meaning while scrambling surface stylistic features through a translation pivot, it functions as a paraphrase-attack probe. We report the same detection number under both interpretations.

3.4 Evaluation

The trained detector is applied to each generator’s 260-article set and to test-split Human articles. We report the percentage classified as AI, mean and median P(machine), hooked-consonant density per 1,000 characters, and digit density. For Aya specifically, we additionally compare detection on output produced from mojibake-corrupted headlines (our first generation run) vs. output produced from Unicode-repaired headlines, as an ablation of input quality.

To test whether our cross-generator finding could be an artifact of data contamination (generators reproducing memorised human source text), we run three further analyses, all reported in Section 4.4. We measure 13-gram token overlap and longest common token substring between each generated article and its paired human source, with Unicode normalisation applied to both sides; we re-evaluate detection on a decontaminated subset that drops every generated article sharing a 13-gram with its source; and we measure output-distribution peakedness by generating multiple samples per headline and computing their pairwise edit-distance diversity, following Dong et al. (2024).

3.5 Native-speaker validation

Two Hausa-speaking colleagues reviewed two artefacts: the eight false-positive Human articles from Section 4.2 and a 30-article random sample of NLLB outputs from Section 4.3. They classified

Seed	Acc	F1	Prec	Rec
42	0.985	0.985	0.970	1.000
1	0.996	0.996	0.992	1.000
7	0.994	0.994	0.989	1.000
123	0.994	0.994	0.989	1.000
Mean $\pm$ std	0.992 $\pm$ 0.005	0.992 $\pm$ 0.005	0.985 $\pm$ 0.010	1.000 $\pm$ 0.000
Sani et al. (2025)	0.9923	0.9921	–	–

Table 1: Replication of Sani et al. (2025) for the AfroXLMR-large baseline across four seeds.

the first set by register without access to the detector’s predictions. For the second set, they rated fluency on a 1–5 scale and judged whether content was preserved. The two reviewers were given identical spreadsheets and worked independently.

4 Results

4.1 Replication

Table 1 reports per-seed and aggregate test metrics on the headline-grouped split.

Our four-seed mean F1 of 0.9924 matches the paper’s reported 0.9921 within run-to-run variance.

Two things about Table 1 are easy to miss. Run-to-run variance is small but real: seeds 42 and 1 differ by 1.1 F1 points, and seed 42 happens to be the worst draw. And recall on the AI class is 1.000 at every seed. The detector never misses a Gemini 2.0 Flash article. Whatever varies between runs, it is entirely on the precision side, in how many Human articles get falsely flagged (between 3 and 8 per run, out of 260).

4.2 What does the detector use? Error analysis on false positives

We examine the eight false-positive Human articles from the seed-42 run. Two Hausa-speaking colleagues independently reviewed each article without access to the detector’s predictions, and both judged all eight to be natural Hausa journalism. So the errors are not scraped boilerplate or degraded text slipping through; the detector is misclassifying genuine human-written news.

The clearest correlate of misclassification is the absence of numerical content. Six of the eight false positives contain zero digit characters, and the median digit-character fraction is 0.000 in the false positives vs. 0.006 in correctly classified Humans (Table 2). Hard-news writing routinely carries dates, ages, casualty counts, and percentages;

the digit-sparse Human articles evidently look more machine-like to the detector. This does not explain everything, though. Two of the flagged articles are statistic-dense and were misclassified anyway.

We had also expected the false positives to concentrate in non-news registers (opinion, feature, cultural commentary), on the intuition that the training data is overwhelmingly hard news (53.5% politics). The native-speaker review did not support this cleanly. While a bylined opinion essay, a celebrity-feud feature, and a current-affairs programme summary were judged opinion or feature, both reviewers classified other false positives (a desertification-policy report, a diabetes and sugar-tax report, and a security-chiefs meeting) as straight news, and the two reviewers did not fully agree on the borderline cases. We therefore do not claim a register effect: the misclassified set spans both non-news and unambiguous hard-news writing, so register does not characterise the errors. Per-article register labels from both reviewers are in Appendix E.

We caution that this analysis is on the seed-42 checkpoint, which had the largest false-positive set (8) among our four seeds. Seeds 1, 7, and 123 each produced only 3 false positives. Confirming the digit-sparsity pattern on those checkpoints is left for future work.

We also asked whether orthographic density explains the false positives. The intuitive story would be that the detector flags Human articles with few hooked consonants as machine-like. Our data say otherwise. Median hooked-consonant density per 1,000 characters is 14.87 in the false positives vs. 8.09 in correctly classified Humans, so the misclassified articles actually have *more* Hausa-specific orthography than the ones the detector gets right. A one-sided Mann-Whitney U test ($FP < TN$) gives $p = 0.96$. At least where its errors are concerned, the detector is not applying a “low Hausa orthography means machine” rule.

Orthographic density does play a role in the AI class, however. On the test-split AI articles, median hooked-consonant density is 0.0 per 1,000 characters: more than half of Gemini-generated articles contain zero hooked consonants. The signal is about presence, not density, and on the training distribution it is strong enough to explain why test recall never drops below 1.000. We return to this in Section 4.3, where the cross-generator data narrow down what the detector can actually be using.

Feature	FP (n=8)	TN (n=252)
Median digit fraction	0.000	0.006
Zero-digit articles	6 (75%)	21 (8%)
Median hooked/1k	14.87	8.09
Natural Hausa (both reviewers)	8 (100%)	not anno.

Table 2: Feature comparison of false-positive vs. correctly classified Human articles (seed 42). The false positives have *higher* hooked-consonant density than correctly classified Humans, refuting an orthographic-sparsity explanation.

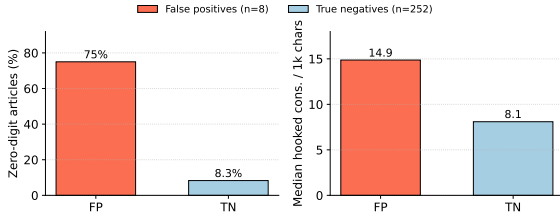


Figure 1: False-positive vs. true-negative Human articles (seed 42): share of zero-digit articles and median hooked-consonant density. The misclassified articles are digit-sparse but orthographically *richer* than correctly classified Humans.

4.3 Cross-generator evaluation

Table 3 and Figure 2 report detector behaviour on six non-training generators, plus the in-distribution training generator and the Human reference. Generators are sorted by detection rate.

Three patterns stand out.

Detection rates span the full range. Across the six non-training generators we see everything from 0% to 100%. The detector generalises to some out-of-distribution generators and fails on others. The English-language cross-generator literature has mostly reported a binary picture, “single-generator detectors do not transfer” (Dugan et al., 2024; Wang et al., 2024); in Hausa, at least for this detector, the reality is messier.

Google-family models are detected at near-perfect rates. Both Google-family models in our test set, Gemma 4 E4B and Gemma 4 E2B, are detected at 100.0% and 98.9% respectively, with mean $P(\text{machine})$ of 1.000 and 0.989. No other tested non-training generator exceeds 21%. The simplest hypothesis is that the detector has learned features shared across Google’s LLM family, plausibly arising from common training data, tokenizer family, instruction-tuning style, or RLHF templates, rather than features general to machine-written Hausa. Models from other families (Co-

here/Aya, Meta/Llama, Alibaba/Qwen, Meta translation/NLLB) lack these features and are missed at rates from 79% to 100%. Across all three Google models in our experiment (the training-distribution Gemini 2.0 Flash plus two non-training Gemma variants of different sizes), detection is consistently above 98% with mean $P(\text{machine})$ above 0.98. Across all four non-Google generators, detection is at or below 20.8% with mean $P(\text{machine})$ at or below 0.21. The gap is large enough that we doubt the pattern is incidental.

Orthographic density does not explain the detection signal. Hausa-specific hooked-consonant density varies substantially across our generators (full per-generator statistics in Appendix D). Aya produces almost no hooked consonants (0.5 per 1,000 characters under mojibake input, 1.5 under repaired input). Gemma 4 produces more than Humans (10.3 to 12.1 per 1,000 characters across the two Gemma variants vs. 7.9 for Humans). The three Google models in our experiment span hooked-consonant densities from 5.2 (Gemini 2.0 Flash) to 12.1 (Gemma 4 E2B), a $2.3\times$ range, yet all three are detected at rates above 98% with maximum confidence. That spread leaves no room for a “machine-generated Hausa lacks hooked consonants” explanation. Hooked-consonant density was a correlate within the training distribution; it is not what the detector learned.

The Aya input-quality ablation makes the same point from a different angle. With mojibake-corrupted headlines, Aya output averages 0.5 hooked consonants per 1,000 characters, with 75% of articles containing zero. With Unicode-repaired headlines, output averages 1.5 hooked consonants per 1,000 characters with 58% zero-hook articles. Cleaning the input triples Aya’s orthographic richness, and the detector does not react at all: 0% called AI in both conditions, mean $P(\text{machine})$ below 0.001. Whatever the detector reads in a generator’s output, it is not the surface orthography.

Aya’s output also deserves a qualitative note. Its Hausa is often partially code-switched into Somali, which a Hausa-speaking reviewer flagged as unfluent. The Aya generation with the highest $P(\text{machine})$ in our run (article #49, $P = 5.3 \times 10^{-5}$) contains explicit Somali function words. The detector still rates it 99.99% Human. The detector is willing to classify text that is not even fluent Hausa as Human, provided it lacks the Gemini-family signature.

Conversely, the Gemma 4 output we self-

Generator	Family	% AI	95% CI	Mean P(machine)	Hooked/1k
Aya-Expanse 8B (mojibake input)	Cohere	0.0	[0.0, 1.5]	< 0.001	0.50
Aya-Expanse 8B (repaired input)	Cohere	0.0	[0.0, 1.5]	< 0.001	1.53
Llama 3.1 8B	Meta	1.2	[0.4, 3.3]	0.013	5.18
Qwen 3 8B	Alibaba	10.0	[6.9, 14.2]	0.108	4.79
NLLB-200 round-trip	Meta (translation)	20.8	[16.3, 26.1]	0.210	3.82
Gemma 4 E2B	Google	98.9	[96.7, 99.6]	0.989	12.09
Gemma 4 E4B	Google	100.0	[98.5, 100.0]	1.000	10.33
Gemini 2.0 Flash	Google (training)	100.0	[98.5, 100.0]	1.000	5.23
Human (test split)	—	3.1	[1.6, 6.0]	0.031	7.94

Table 3: Cross-generator detector behaviour (n=260 each, seed-42 checkpoint). Intervals are Wilson 95% confidence intervals for the percentage classified as AI.

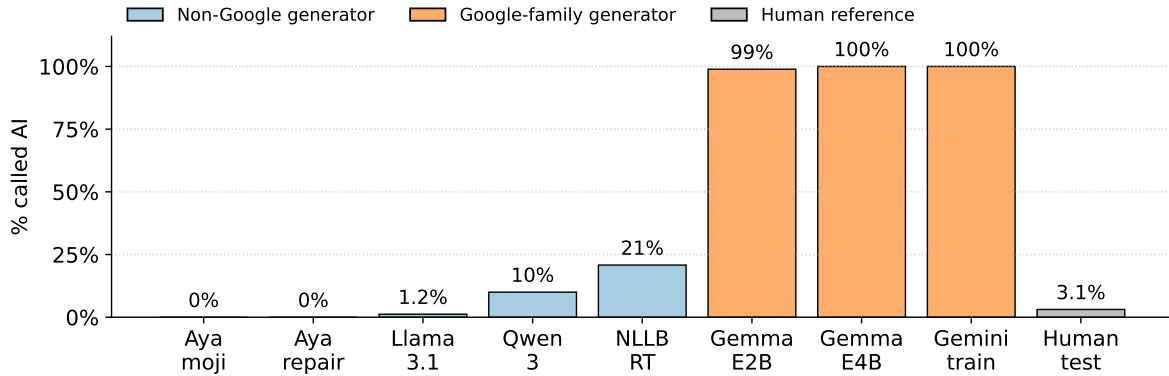


Figure 2: Detection rate by generator (n=260 each, seed-42 checkpoint). All three Google models are detected at or above 99%; no non-Google generator exceeds 21%.

inspected (10 random samples from the 260-article set) is fluent, on-topic Hausa news prose with proper hooked-consonant orthography throughout (mean 299 words per article, hooked-consonant density of 10.3 per 1,000 characters, higher than the Human reference at 7.9). We did notice recurring article-opening templates, “*An tattara labarin*” and “*An bayar da labari*” among them, the kind of stylistic uniformity a detector could plausibly latch onto. We saw no language drift, repetition loops, or refusals. So the 100% detection of Gemma 4 cannot be blamed on degraded output; the detector is confidently flagging genuine, fluent Hausa that happens to carry Google-family stylistic features. Appendix C gives qualitative examples from each generator.

4.4 Is the family signature a contamination artifact?

The human articles in the Sani et al. (2025) corpus were scraped from public Hausa news sites, so they may appear in the pretraining data of the generators we test. If a generator had memorised a human source article, its “machine-generated” out-

put could carry verbatim spans of that human text, and the detector might then be responding to regurgitated human writing rather than to a genuine generator signature. We run three analyses to test this.

First, we measure surface overlap between each generated article and its paired human source, using 13-gram token overlap (the convention of Brown et al., 2020) and longest common token substring, with both sides Unicode-normalised so that mojibake does not artificially depress the counts. As a control, we also measure each generated article’s maximum overlap against a random sample of *other* human articles, to separate topical overlap (shared headline, shared topic words) from source-specific copying. Table 4 reports the result. Source overlap is low in absolute terms for every generator, and, more importantly for the family-signature claim, it is not higher for Google models. Gemma 4 E4B has the *lowest* source overlap of any generator (mean 13-gram Jaccard 0.0002, 4.6% of articles sharing any 13-gram) yet is detected at 100%. Qwen 3 has by far the *highest* overlap

Generator	src Jac.	ctrl Jac.	% w/ 13-gram
Aya (mojibake)	0.0027	0.0000	23.1
Aya (repaired)	0.0025	0.0000	23.5
Llama 3.1 8B	0.0017	0.0000	15.0
Qwen 3 8B	0.0278	0.0001	76.2
NLLB round-trip	0.0010	0.0002	23.1
Gemma 4 E2B	0.0087	0.0000	51.2
Gemma 4 E4B	0.0002	0.0000	4.6

Table 4: Surface overlap between generated articles and human sources. **src Jac.** is mean 13-gram Jaccard against the paired human source; **ctrl Jac.** is mean maximum 13-gram Jaccard against random other human articles; **% w/ 13-gram** is the fraction of articles sharing at least one 13-gram with their source. Google’s Gemma 4 E4B has the lowest source overlap yet the highest detection; Qwen has the highest overlap yet near-floor detection.

(0.028, 76.2% sharing a 13-gram) yet is detected at only 10%. If anything, detection runs in the opposite direction from overlap: the generator that copies least is caught most.

Second, we re-run the full cross-generator evaluation on a decontaminated subset, dropping every generated article that shares at least one 13-gram with its paired human source and recomputing the detection rate on what remains (Table 5). This removes 51% of Gemma 4 E2B’s articles and 76% of Qwen’s, a severe filter. The headline finding is unchanged. Both Google models remain at or above 99% on the clean subset (Gemma 4 E2B in fact rises slightly, from 98.85% to 99.21%, as its most source-similar articles are removed), while no non-Google generator approaches Google territory. Qwen’s detection *falls* from 10.0% to 4.8% under decontamination, so what little detection Qwen attracted came partly from its most headline-echoing articles. Where overlap mattered at all, it was proping up the score of a weakly detected model; it cannot be the source of the Google result.

The one non-trivial upward shift is NLLB (20.8% to 24.0%). This is expected and mild: NLLB output is a back-translation of the human source, so its highest-overlap articles are the ones that stayed closest to the human original, and removing them leaves a more drifted, more machine-like remainder. That fits the paraphrase-probe reading of the NLLB condition (Section 4.5).

Third, we test whether the Google models might be detected because their output is memorisation-like in a way the surface-overlap analyses would miss, namely low-diversity or templated generation.

Generator	full %	clean %	Δ
Aya (repaired)	0.0	0.0	0.00
Aya (mojibake)	0.0	0.0	0.00
Llama 3.1 8B	1.2	1.4	+0.20
Qwen 3 8B	10.0	4.8	−5.16
NLLB round-trip	20.8	24.0	+3.23
Gemma 4 E2B	98.9	99.2	+0.37
Gemma 4 E4B	100.0	100.0	0.00

Table 5: Detection rate before (**full**) and after (**clean**) removing every generated article that shares a 13-gram with its human source. The family signature survives decontamination; the two Google models stay at $\geq 99\%$ while non-Google models stay low.

Generator	peaked frac.	mean pair ED
Qwen 3 8B	0.0098	0.780
Aya-Expanse 8B	0.0000	0.844
Gemma 4 E2B	0.0000	0.824
Gemma 4 E4B	0.0000	0.892
Llama 3.1 8B	0.0000	0.905

Table 6: Output-distribution peakedness (15 samples $\times$ 40 headlines per generator). **peaked frac.** is the fraction of sample pairs with normalised edit distance ≤ 0.05 ; **mean pair ED** is mean pairwise normalised edit distance (higher = more diverse). Both Google models produce zero near-duplicate pairs and highly diverse output, ruling out a memorisation explanation for their detection.

Following the output-distribution intuition of Dong et al. (2024), we generate 15 independent samples per headline for 40 headlines from each local generator and measure how self-similar those samples are (sampling protocol in Appendix F). We report two quantities per generator: the fraction of sample pairs whose normalised token edit distance is at most 0.05 (a peakedness measure; higher means more near-duplicate generation), and the mean pairwise normalised edit distance (higher means more diverse). Table 6 shows the result. The two Google models are not peaked at all: both Gemma 4 variants produce zero near-duplicate sample pairs, and their samples are as diverse as, or more diverse than, the non-Google generators (mean pairwise edit distance 0.82 and 0.89, against 0.78 for Qwen and 0.84 for Aya). If the detector were keying on memorisation-like, low-diversity Google output, we would expect the Google models to be the most peaked; they are among the least. Whatever the detector reads in Google output, it persists across samples that differ almost completely from one another.

None of the three checks leaves room for a contamination account of the Google-family gradient. The generators do not reproduce their human sources (Tables 4 and 5), and the Google output the detector flags is diverse, with no sign of memorisation (Table 6).

We cannot rule out a distinct contamination pathway on the *detector* side: AfroXLMR-large was itself pretrained on Hausa web text and may have seen the test-set human articles. However, this cannot explain the central finding, because detector-side memorisation of *human* articles offers no mechanism for the detector to fire on Gemma output while ignoring Aya, Llama, and Qwen output, all of which are machine-generated articles the detector never saw during its own fine-tuning. We discuss this limitation further in Section 5.1.

4.5 Paraphrase robustness via NLLB round-trip

The NLLB round-trip also serves as a paraphrase-attack probe. Translation through English preserves meaning while scrambling surface stylistic features. On NLLB-paraphrased Hausa articles, the detector classifies 20.8% as AI, compared to 100% on the original Gemini output. The detector retains partial signal after this transformation but loses approximately four-fifths of it. In a deployment context, an adversary can evade detection on roughly 80% of articles by running LLM output through a translation round-trip.

To check that the NLLB output is fluent enough to count as a meaningful paraphrase rather than degraded noise, two Hausa-speaking annotators rated a 30-article sample of round-trip outputs for fluency (1–5) and content preservation. The two reviewers diverged in strictness. A lenient reviewer rated most outputs 4–5 and judged nearly all as natural Hausa with preserved content; a stricter reviewer rated roughly a third of the sample at 3 or below and flagged hallucinated named entities and topic drift on the harder articles (for example, a desertification agency rendered as a “cooking agency”, and a post-partum health column drifting into unrelated material). Both reviewers converged on the same qualitative picture: the round-trip output reads as surface-fluent Hausa, but harder source articles accumulate content drift and occasional invented entities. That is close to the profile a paraphrase probe needs, though plainly an imperfect one. Per-article ratings are in Appendix E.

4.6 Real-world false-positive rate

On the test-split Human articles, the detector classifies 3.1% as AI. For deployment this is the number that matters: in a moderation pipeline, roughly 3 of every 100 real Hausa articles would be falsely flagged, and the error analysis in Section 4.2 suggests digit-sparse articles are the most exposed.

5 Discussion

Three implications follow from our results.

First, **generator-family specificity is the central deployment risk**. Our cross-generator gradient shows the detector reliably catches text from one model family (Google) and largely misses text from other families. Practitioners deploying detectors trained on a single generator’s output should expect family-specific behaviour and either retrain when the underlying generator changes or train on a broader generator mix from the outset. This empirically confirms the concern Sani et al. (2025) raised in their own limitations section.

Second, **the adversary’s evasion path is trivial**. Anyone wanting to publish AI-written Hausa text without detection can use Aya-Expanse, Llama, Qwen, or an NLLB-translated rewrite of an English source. All evade the detector at rates above 79%.

Third, **the false positives matter operationally**. A 3.1% Human FP rate sounds tolerable in isolation, but the flagged articles are real journalism, and digit-sparse pieces (no dates, ages, or statistics) are hit hardest. A journalist wrongly flagged as a machine has little practical recourse, so false-positive behaviour deserves monitoring in its own right, separately from aggregate accuracy.

5.1 Limitations

Our cross-generator evaluation tests six generators. Three of them are Google models (Gemini 2.0 Flash, the training generator, plus Gemma 4 E4B and E2B). While the Google-family detection pattern across these three is sharp and consistent (all detected at $\geq 98\%$, with maximum confidence on two of three), we acknowledge the family-signature claim would be further strengthened by an additional non-Google MoE model or a separate Google model from a different architectural lineage. The current evidence is consistent with the family-signature hypothesis but does not exclude alternative explanations such as multilingual-training-data overlap or shared instruction-tuning corpora.

We do not perform probing experiments to identify the specific features the detector keys on. Approaches such as LIME, SHAP, or attention-based attribution would help pinpoint whether the Gemini-family signal arises from specific tokens, n-grams, sentence patterns, or higher-level stylistic features. We leave this for future work.

We evaluate on a single language. The orthographic-density finding raises questions about whether similar patterns arise for Yoruba (tone marks), Igbo (subdot diacritics), and other African languages with non-ASCII orthography. Multi-language replication is an obvious next step.

Cross-generator evaluation is based on a single seed-42 checkpoint. Multi-seed evaluation across the cross-generator data would confirm whether 0% on Aya and 100% on Gemma 4 hold across initialisations. Given that AI-class recall was 1.000 at every seed on the training distribution, we expect the cross-generator pattern to replicate, but we have not verified this.

The training-distribution model (Gemini 2.0 Flash) was deprecated by Google in March 2026, which means within-family transfer between Gemini 2.0 and Gemini 2.5 Flash cannot be cleanly tested from a research-budget free tier. Future work with paid API access could close this gap.

On contamination, our analyses in Section 4.4 address the generator-side pathway (generators regurgitating memorised human source text) and show the family signature survives it. We cannot directly test the detector-side pathway: AfroXLMR-large was pretrained on Hausa web text and may have encountered the test-set human articles before our fine-tuning. We argue this does not threaten the central finding, since memorisation of human articles gives the detector no mechanism to separate Gemma output from Aya, Llama, or Qwen output, which are all machine articles unseen at fine-tuning time. A full test would require retraining the encoder on a corpus with the evaluation articles held out, which is beyond our compute budget.

5.2 Future work

- Probing the learned features through attribution analysis on the trained checkpoint to identify the actual Google-family signature.
- Multi-language replication on Yoruba and Igbo to test whether orthographic and family-signature findings generalise.
- A direct LLM-paraphrase attack with quality

controls, complementing the NLLB round-trip evidence.

- Additional Google-family generators (Gemma 4 26B, paid-tier Gemini 2.5 runs) to strengthen the family-signature claim.
- Re-running the error analysis on seeds 1/7/123 to test whether the digit-sparsity pattern replicates across initialisations.
- Extending the dataset to non-news domains and additional generators, consistent with the future work [Sani et al. \(2025\)](#) outlined in their original paper.

6 Conclusion

We confirm [Sani et al. \(2025\)](#)’s AfroXLMR-large detector replicates cleanly across four random seeds, achieving F1 of 0.992 ± 0.005 . Recall on the AI class is perfect at every seed. All errors are false positives on genuine human-written news, with the absence of numerical content the clearest correlate of misclassification. Across six non-training generators, detection rates span from 0% to 100%; the two non-training Google-family models in our test set are both detected at rates above 98% with maximum confidence, matching the training-distribution rate, while no non-Google generator exceeds 21%. On our evidence the detector is better described as a Gemini-family-output classifier than as a general Hausa MGT detector, and practitioners should expect single-generator detectors to behave in this family-specific way. Our work addresses the single-generator limitation [Sani et al. \(2025\)](#) acknowledged in their original paper, and we hope our cross-generator results inform the design of future Hausa MGT detection benchmarks.

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A Generator prompts

All open generators received the same chat-template prompt structure.

System message:

You are a professional Hausa journalist writing for a major Nigerian newspaper. You write in fluent, natural Hausa using standard orthography including the hooked consonants *ɓ*, *ɗ*, *ɓ̌*, *y̌* where appropriate. You write only in Hausa, never mixing other languages.

User message template:

Write a complete Hausa news article based on the following headline. The article should be between 300 and 500 words. Write only the article, no preamble, no translation, no explanation.

Headline: {headline}

The {headline} placeholder was filled with the headline of a Human article from the test split, after passing it through ftfy to repair Unicode mojibake. For thinking-capable models (Qwen 3, Gemma 4), the thinking budget was set to zero via the chat-template `enable_thinking=False` flag.

For the NLLB round-trip generator, no chat template was used. The Human article was translated Hausa → English with source tag `hau_Latn` and target tag `eng_Latn`, then back English → Hausa with the tags reversed. Long articles were chunked at 800-character boundaries.

Sampling parameters for all open generators: temperature 0.8, top-p 0.95, max output tokens 600 (1,500 for thinking-capable models, with thinking disabled).

B Mojibake in the released dataset

The released CSV contains Unicode corruption of Hausa hooked consonants in approximately 9% of articles. Examples (raw → repaired): `karÃfÃ,Ã,Ã’uwa` → `karɓuwa`; `ÃfÃ,Ã,Ã’oÃfÃ,Ã,Ã’ari` → `ɓɔɓari`; `ÃfÃ,Ã,Ã’arshen` → `ɓarshen`. The pattern is consistent with UTF-8 bytes being misinterpreted

as Latin-1 and re-encoded, a common scraping pipeline error.

We did not modify the dataset for our training runs; we trained on the data as released. For cross-generator experiments, we passed headlines through ftfy before sending them to the generators, so the generators received valid Hausa orthography.

C Examples of cross-generator output

All generators below were given the same Hausa input headline about a wedding-day attack in Myanmar (headline_id 1383 in our test split):

Students’ Revolutionary Force —Ranar daurin auren Hnin Si a farkon watan Yuni, wanda ya kamata ya zama ranar farin ciki a ɓauiyen Mataw da ke yankin Sagaing na Myanmar, ya rikiɗe zuwa bala’i.

C.1 Aya-Expanse 8B

Aya routinely produced output containing material in non-Hausa languages, most often Somali. The Hausa-speaking reviewer characterised it as code-switched and not fluent. From an Adam Zango profile headline:

*Adam Zango Fitaccen, kuna da kalla da Kannywood, **ayaa ka soo baxay** da amma n’ita aikinsa alhalarin kallu da abokon hira. Wanaa 20 sanno da soo baxay da aikinsa, ta’an da aikinsa akochan hira da kallu, **waa inuu** abokon hira da Adam Zango Fitaccen aikinsa ka soo baxay.*

The bolded function words (*ayaa ka soo baxay*, *waa inuu*) are Somali, not Hausa. The detector classified this article as Human with $P(\text{machine}) = 1.4 \times 10^{-5}$, near the median for the full Aya generation set.

C.2 Llama 3.1 8B

Llama produced grammatically reasonable Hausa with attempted journalistic detail. On the wedding-bombing headline:

Ranar daurin auren Hnin Si, wanda aka fi sani da ‘Hnin Si’s Wedding Day’, ya bayyana ne a farkon watan Yuni, wanda ya kai laushi a ɓauiyen Mataw, cikin yankin Sagaing na Myanmar. Hnin Si, ‘yar shekara 25, ta auri ɗan’uwan auren ta, Linn Htwe, a ranar 11 ga watan Yuni...

Names and ages are partly invented (the real headline does not name the groom), but the prose is on-topic and reads as Hausa news. The detector classified this article as Human ($P(\text{machine}) = 0.013$, near the Llama mean).

C.3 Qwen 3 8B

Qwen 3 frequently entered repetition loops, repeating a phrase or sentence with minor variations until the token budget was exhausted. On the wedding-bombing headline:

*Ranar daurin auren Hnin Si, wanda ya kamata ya zama ranar farin ciki a wauyen Mataw, da ke yankin Sagaing na Myanmar, ya rikiƙe zuwa bala'i. **Daga kuma, ranar daurin auren Hnin Si, wanda ya kamata ya zama ranar farin ciki a wauyen Mataw, da ke yankin Sagaing na Myanmar, ya rikiƙe zuwa bala'i. Kuma, ranar daurin auren Hnin Si...***

The repeated clauses are bolded. Despite the repetition, the detector classified 10% of Qwen 3 outputs as AI, the second-highest non-Google rate after NLLB.

C.4 NLLB-200 round-trip (paraphrase)

The NLLB round-trip preserves the topic but produces characteristic translation artefacts. On the wedding-bombing headline:

Hsin Si ta ce yayin da kusan abokai da dangi 100 ke taruwa don bikin bikin aurenta, jirgin sama ya jefa bama-bamai ba zato ba tsammani, ya kashe mutane 33, ciki har da mata da yara. “Bayan da muka yi aure, mun fara daukar hotuna tare da iyalai da abokai, kawai don jin bam sun fara tsakaninmu yayin da Eaya ta fashe kusa da gidan amarya”. Yankin Sagaing, tare da Mandalay da Magway...

This is a back-translation of the source Human article. The named entity “Hnin Si” became “Hsin Si” through the pivot. The detector classified this article as AI ($P(\text{machine}) = 0.45$). 21% of NLLB outputs were classified as AI overall.

C.5 Gemma 4 E4B

Gemma 4 produced fluent, on-topic Hausa with consistent stylistic patterns. On the wedding-bombing headline:

An tattara labarin daga yankin Sagaing a Myanmar inda an samu wani abin mamaki da kuma walubale da ya faru a kan wani taron tarihi mai muhimmanci. wungiya ta kan malamai da dalibai, da aka fi sani da sunan Hukumar Daidaito ta dalibai (Students' Revolutionary Force), ta shirya wani taro don fara bikin aure na Hnin Si a farkon watan Yuni...

The bolded opening (“An tattara labarin” — “the news has been gathered”) recurs across many Gemma 4 generations and is one of the recurring stylistic templates we noted in Section 4.3. The detector classified this article as AI with $P(\text{machine}) \approx 1.000$. 100% of Gemma 4 E4B outputs were classified as AI.

C.6 Gemma 4 E2B

The smaller Gemma 4 variant produced similar Hausa news prose with some content drift. On the wedding-bombing headline:

Wata wungiya ta dalibai ta yi tashin hankali a wauyen Mataw na yankin Sagaing na Myanmar a farkon watan Yuni, wanda ya yi tsanani da gaba daya da ya haifar da wani tsanani mai tsanani...

Note the awkward phrase repetition (“tsanani da gaba daya da ya haifar da wani tsanani mai tsanani”). The article preserves the topic but loses the wedding framing entirely. The detector classified this article as AI with $P(\text{machine}) \approx 1.000$. 98.9% of Gemma 4 E2B outputs were classified as AI.

D Hooked-consonant density distributions

Generator	Mean	Med.	% zero
Human (test split)	7.94	8.58	27.7
Gemini 2.0 Flash	5.23	0.00	58.9
Aya 8B (mojibake)	0.50	0.00	75.0
Aya 8B (repaired)	1.53	0.00	58.5
Llama 3.1 8B	5.18	2.60	26.2
Qwen 3 8B	4.79	0.25	48.9
Gemma 4 E2B	12.09	10.83	0.8
Gemma 4 E4B	10.33	9.82	0.0
NLLB-200 round-trip	3.82	3.38	18.5

Table 7: Hooked-consonant statistics per generator (per 1,000 characters).

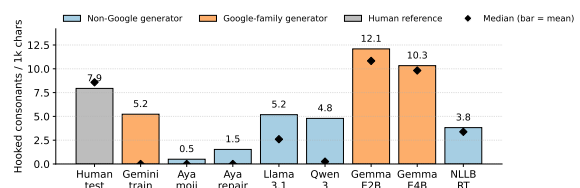


Figure 3: Mean and median hooked-consonant density by generator. The Google-family generators vary widely in orthographic density while remaining highly detectable, whereas some non-Google generators have similar densities to Gemini but much lower detection rates.

Both Gemma 4 variants have non-zero hooked-consonant density in virtually all articles, exceeding the Human reference. Figure 3 shows that the three Google models span 5.2 to 12.1 hooks per 1,000 characters but are all detected at $\geq 98\%$, refuting the hypothesis that orthographic density is the detection signal.

E Native-speaker validation protocol

Two Hausa-speaking colleagues each received an identical Excel spreadsheet with two sheets and worked independently. The first sheet contained the eight false-positive Human articles with columns for register classification (news, opinion, feature, cultural, other), a yes/no judgement on whether the article reads as a straight news report rather than opinion or feature, a yes/no judgement on whether the language is natural Hausa journalism, and free-text comments. Reviewers were not told which articles the detector had misclassified or what hypothesis was under test.

Table 8 summarises the register classifications. Both reviewers judged all eight articles to be natural Hausa journalism (8/8 “natural Hausa = yes”), confirming the false positives are genuine human writing rather than scraped noise. On register, the reviewers diverged: one assigned single-register labels (four opinion/feature/cultural, three news, one mixed), while the other often gave compound labels such as “news, feature, as well as cultural”. Because the two agreed that several false positives (desertification policy, diabetes/sugar-tax report, security-chiefs meeting) are straight news, we do not treat register as characterising the errors; the digit-sparsity correlate in Section 4.2 is the finding the data supports.

FP	Topic	R1	R2
1	Tailoring/fashion	news/feat/cult	cultural
2	Desertification	news	news
3	Diabetes report	news	news
4	SA election	news	opinion
5	Musician feud	news/feat/opin	feature
6	Tax-bill TV prog.	news	opinion
7	Security meeting	news	news
8	Africa-image essay	news/opinion	opinion

Table 8: Native-speaker register classifications of the eight false positives. R1 and R2 are the two independent reviewers. Both rated all eight as natural Hausa journalism.

The second sheet contained 30 randomly sampled NLLB round-trip outputs paired with their source Human articles, with columns for fluency (1–5), a yes/no judgement on whether a Hausa reader would accept the text as written by a Hausa journalist, a yes/no judgement on whether the translation preserved the source article’s key facts, and free-text comments. The two reviewers diverged in strictness. The lenient reviewer gave a mean fluency near 4.6, with natural-Hausa = yes and

content-preserved = yes on almost all 30 items. The strict reviewer gave a mean fluency near 3.5 and flagged content problems on roughly a third of the sample, including hallucinated named entities and topic drift (for example, NLLB_12, where the “Great Green Wall” agency became a “cooking agency”, and NLLB_15, where a post-partum health column drifted into unrelated material). A website-boilerplate item (NLLB_13: “Wikki Times Media... All Rights Reserved”) made the difference plain: the strict reviewer scored it 1 (“this has no meaning”), the lenient reviewer 5. Both reviewers’ free-text comments recurred on the same theme, summarised by one as “a little hallucination as well as grammatical errors”. Both ended up in the same place: NLLB round-trip output is surface-fluent Hausa with measurable content drift on harder source articles, which is what makes it usable as a paraphrase probe.

F Software and reproducibility

All experiments were run on a single Ubuntu 24.04 workstation with an NVIDIA RTX 3090 (24 GB VRAM) and 64 GB RAM. Python 3.11, PyTorch 2.11.0+cu130, transformers 5.8.1, vLLM 0.21.0, ftfy 6.1.

Random seeds for the four-seed training runs: 42, 1, 7, 123. The vLLM sampler was set to seed 42 for all generator runs to make the generations reproducible. Hugging Face model commits at time of access can be reconstructed from the dates of our runs (May 2026).

For the peakedness analysis (Table 6), we generated 15 samples per headline for 40 randomly selected test headlines from each local generator, at the same temperature 0.8 and top-p 0.95 used for the main generations, with the vLLM sampler seeded at 42. Peakedness is the fraction of within-headline sample pairs whose token-level normalised edit distance is at most 0.05; diversity is the mean pairwise normalised edit distance.

All scripts used to produce the results in this paper, along with the trained-checkpoint hashes and the cross-generator output CSVs, will be released on publication.